2.1.3 ▶ ▶ Internal Energy

The total internal energy of a thermodynamic system is defined as the total energy contained within it. For an ideal gas, since there are no internal potentials, this energy is simply the total kinetic energy of the gas particles, or

$$U_{int} = N\langle K\rangle \tag{2.12}$$

where N is the number of particles and $\langle K\rangle$ is the number of gas particles. Using the mechanical definition of kinetic energy,

$$\langle K\rangle = \frac{1}{2}m\langle v^2\rangle = \frac{1}{2}mv_{rms}^2$$

Hence, the total internal energy of a monatomic gas is

$$U_{int} = \frac{3}{2}K_BT. \tag{2.13}$$

Alternatively, if we define n to be the number of moles of gas constant and the constant R to be Avogadro's number times k_B, we have

$$U_{int} = \frac{3}{2}(nN_A)k_BT = \frac{3}{2}nRT \tag{2.14}$$

Notice how this result is specifically for *monatomic* gas, and you can think of the equation as the sum of the average kinetic energy in each degree of freedom (x,y,z axis), which gives rise to the factor of 3/2.

$$K_i = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2$$

For diatomic gasses, there are two[5] additional rotational degrees of freedom, namely

$$K_i = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 + \frac{1}{2}I_x\boldsymbol{\omega}_x^2 + \frac{1}{2}I_y\boldsymbol{\omega}_y^2$$

Logically, the total internal energy should be

$$U_{int} = \frac{5}{2}Nk_BT \tag{2.15}$$

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[5]Two, not three, since there is symmetry about two axes. For objects with less symmetry, there will be three rotational axes (Try it! Try to find more than two unique rotational axes for a pencil, then try the same with a smartphone!).